

CSE 4417 IoT Theory

Lecture 03: Sensors

Atanu Shuvam Roy



Lecturer,
Dept. of CSE, ULAB, Dhaka



EMAIL:

atanu.shuvam@ulab.edu.bd



Class Code
bjedchaj



Spring 2026

Sensing

Sensors

- A sensor detects (senses) changes in the ambient conditions or in the state of another device or a system, and forwards or processes this information in a certain manner [1].
- “A device which detects or measures a physical property and records, indicates, or otherwise responds to it” [2].

Oxford Dictionary

- References-

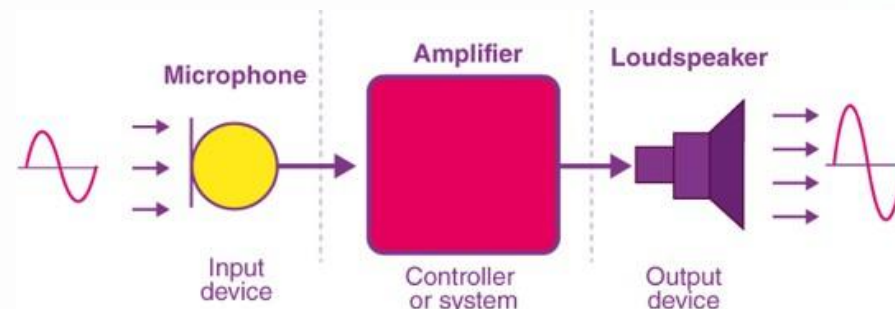
1. <http://www.businessdictionary.com/definition/sensor.html>
2. <https://en.oxforddictionaries.com/definition/sensor>

Sensors

- Sensors perform some input functions by sensing or feeling the physical changes in characteristics of a system in response to a stimuli.
- For example heat is converted to electrical signals in a temperature sensor, or atmospheric pressure is converted to electrical signals in a barometer.

Transducers

- Transducers convert or transduce energy of one kind into another.
- For example, in a sound system, a microphone (input device) converts sound waves into electrical signals for an amplifier to amplify (a process), and a loudspeaker (output device) converts these electrical signals back into sound waves.



Sensor Features

- It is only sensitive to the measured property (e.g., A temperature sensor senses the ambient temperature of a room.)
- It is insensitive to any other property likely to be encountered in its application (e.g., A temperature sensor does not bother about light or pressure while sensing the temperature.)
- It does not influence the measured property (e.g., measuring the temperature does not reduce or increase the temperature).

Sensor Resolution

- The sensor resolution or measurement resolution is the smallest change that can be detected in the quantity that is being measured. The resolution of a sensor with a digital output is usually the numerical resolution of the digital output
- The more is the resolution of a sensor, the more accurate is its precision.
- A sensor's accuracy does not depend upon its resolution.

Sensor Classes

Based on Output

Analog

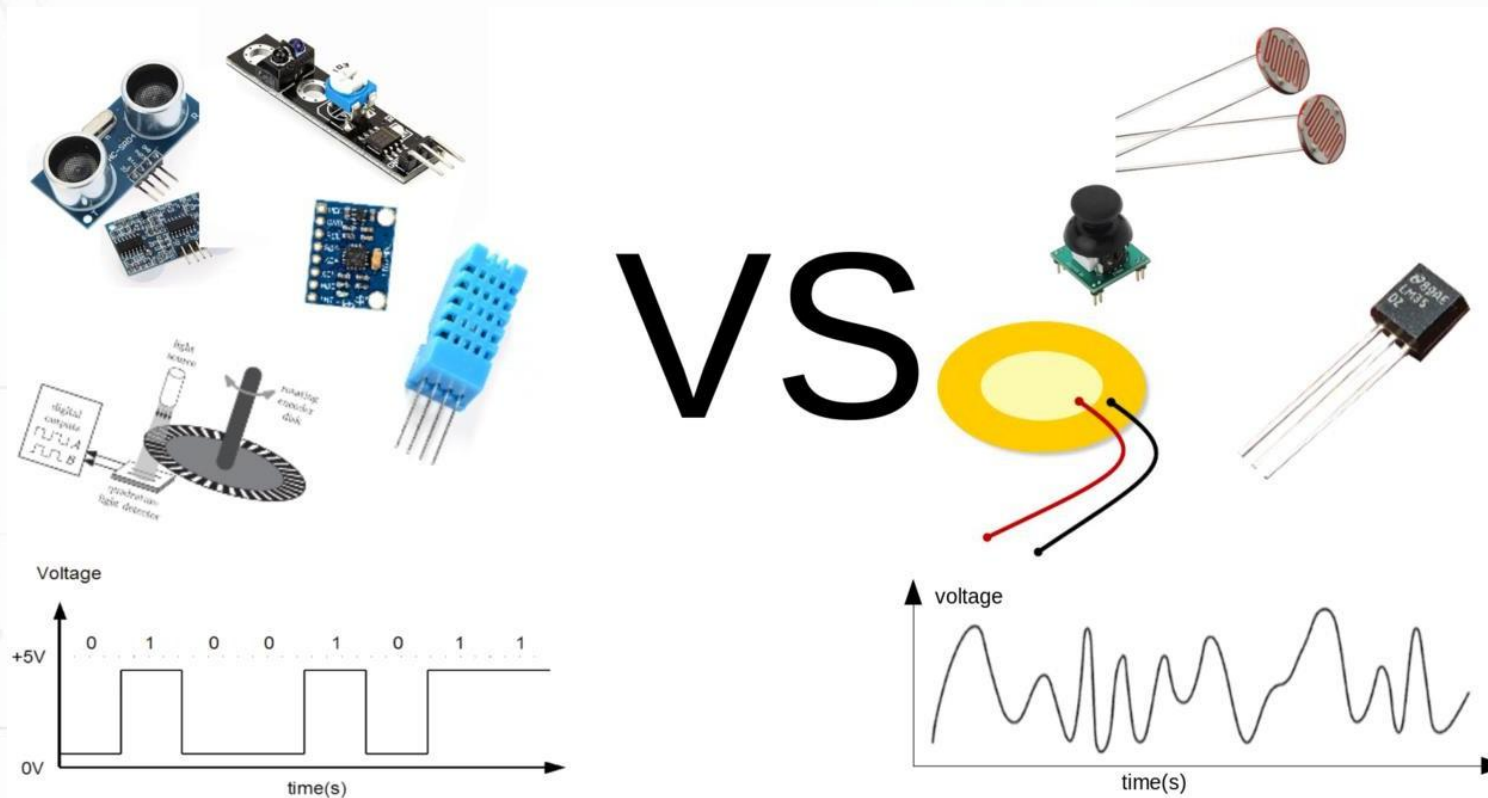
Digital

Based on Data Type

Scalar

Vector

Digital Vs Analog Sensors



Digital Sensors

- Digital Sensors produce **discrete digital output signals** or voltages that are a digital representation of the quantity being measured.
- Digital sensors produce a binary output signal in the form of a logic “1” or a logic “0”, (“ON” or “OFF”).
- Digital signal only produces discrete (non-continuous) values, which may be output as a single “bit” (serial transmission), or by combining the bits to produce a single “byte” output (parallel transmission)

Analog Sensors

- Analog Sensors produce a **continuous output signal or voltage** which is generally proportional to the quantity being measured.
- Physical quantities such as Temperature, Speed, Pressure, Displacement, Strain etc. are all analog quantities as they tend to be continuous in nature.
- For example, the temperature of a liquid can be measured using a thermometer or thermocouple (e.g. in geysers) which continuously responds to temperature changes as the liquid is heated up or cooled down.

Scalar Sensors

- Scalar Sensors produce output signal or voltage which is generally proportional to the **magnitude of the quantity** being measured.
- Physical quantities such as temperature, color, pressure, strain, etc. are all scalar quantities as only their magnitude is sufficient to convey an information.
- For example, the temperature of a room can be measured using a thermometer or thermocouple, which responds to temperature changes irrespective of the orientation of the sensor or its direction.

Vector Sensors

- Vector Sensors produce output signal or voltage which is generally proportional to the **magnitude, direction, as well as the orientation** of the quantity being measured.
- Physical quantities such as sound, image, velocity, acceleration, orientation, etc. are all vector quantities, as only their magnitude is not sufficient to convey the complete information.
- For example, the acceleration of a body can be measured using an accelerometer, which gives the components of acceleration of the body with respect to the x,y,z coordinate axes.

Sensor Types

Light	• Light Dependent resistor • Photo-diode
Temperature	• Thermocouple • Thermistor
Force	• Strain Gauge • Pressure switch
Position	• Potentiometer, Encoders
Speed	• Reflective/ Opto-coupler • Doppler effect sensor
Sound	• Carbon Microphone
Chemical	• Liquid Chemical sensor • Gaseous chemical Sensor

Vector Sensors



Pressure Sensor



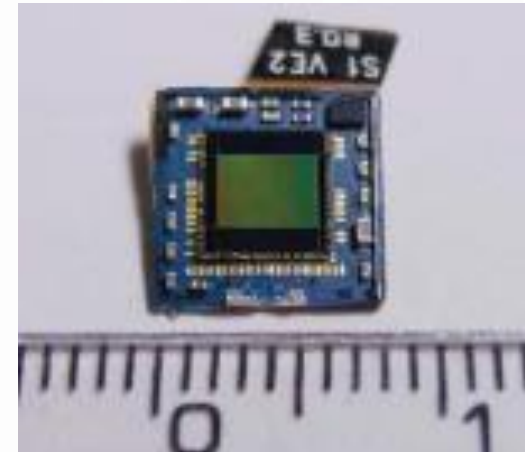
Ultrasonic Distance Sensor



Infrared Motion Sensor



Analog Temperature Sensor



Camera Sensor

Hysteresis Error

- A hysteresis error causes the sensor output value to vary depending on the sensor's previous input values.
- If a sensor's output is different depending on whether a specific input value was reached by increasing or decreasing the input, then the sensor has a hysteresis error.
- **The present reading depends on the past input values.**
- Typically, in analog sensors, magnetic sensors, heating of metal strips.



ULAB

UNIVERSITY OF LIBERAL ARTS
BANGLADESH

Thank You

atanu.Shuvam@ulab.edu.bd

University of Liberal Arts Bangladesh
Department of Computer Science & Engineering